

Formal Definition: Function of Two Variables

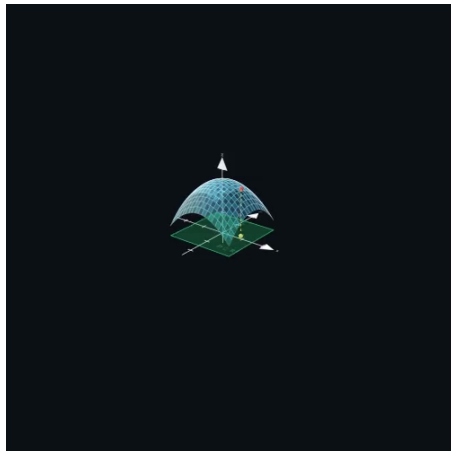
Definition (Function of Two Variables)

Let $z = f(x, y)$ be a function with two variables x and y .

- x and y are the **independent variables**.
- z is the **dependent variable**.
- Unless otherwise specified, the **domain** D is the **largest subset of $\mathbb{R}^2$** on which f is mathematically well-defined.

Geometric Mapping

$$f : D \subseteq \mathbb{R}^2 \longrightarrow \mathbb{R}, \quad (x, y) \longmapsto z$$



Input $(x, y) \in D$ maps vertically to Output $z = f(x, y)$